

A guidance counselor at Lincoln High School in Boston, MA wants to analyze scores on the ACT to increase the average scores for seniors. The counselor gathered the ACT scores for all seniors at the school. The average ACT score in the US is 20.8.

The counselor picked a random sample of 160 senior scores and recorded the information in the table below. Use this data to answer questions 1-5.

Students that scored less than 20.8	Students that scored 20.8 and higher.
104	56

1. What is the proportion of students that scored less than 20.8 on the ACT at Lincoln High School?

$$\frac{104}{160}$$

2. Estimate the probability of a senior in the random sample scoring less than 20.8 on the ACT? Explain your method.

$$\frac{104}{160} = 0.65 = 65\% \quad \text{divide number of seniors in sample who scored less than 20.8 by total in sample.}$$

3. There are 580 seniors at Lincoln High School. Using equivalent fractions, about how many seniors are expected to score less than 20.8 on the ACT (based on the sample data)?

$$\frac{104}{160} \times \frac{3.625}{3.625} = \frac{377}{580} \quad 377 \text{ seniors}$$

4. If you use the percent method to answer question 3, what similarities and differences would you notice? Explain.

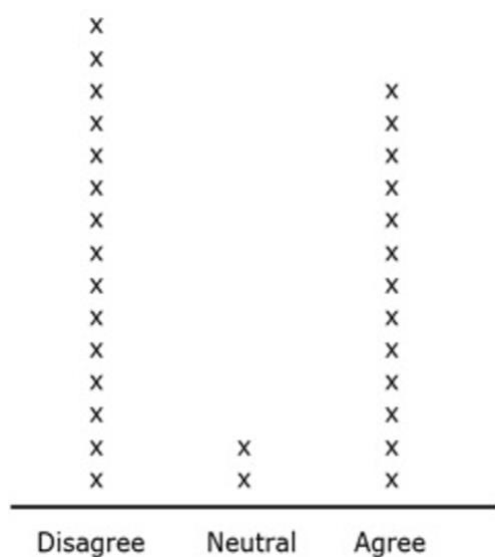
$$0.65 (580) = 377$$

Similarities	Differences
Same answer different method.	Multiply percent by 580.

5. The guidance counselor estimates that 312 seniors will score 20.8 or greater on the ACT sample data. Explain whether this would be considered unreasonable based on the sample data.

$$\frac{56}{160} = 35\% \rightarrow 0.35 (580) = 203 \text{ Therefore 312 seniors are not reasonable since 35\% of 160 is 203, way less than 312.}$$

A middle school in Albany, GA is considering requiring students to wear uniforms for the upcoming school year. The decision was based on proposals by the school advisory committee and PTSA. Parents, teachers, and staff overwhelmingly agreed that wearing uniforms have a positive effect on the school. The school's principal conducted a random survey of 30 students to gauge student opinions on the potential mandatory uniform policy. The results are shown on the line plot below. Use the data to answer questions 6-10.



6. What fraction of students surveyed disagreed on the potential mandatory school uniform policy?

$$\frac{15}{30} = 50\%$$

7. A study conducted by a research group in Nevada indicated that 46% of middle school students disagreed with a mandatory uniform policy. Explain how this compares to the students at the school in Albany that disagree with a mandatory uniform policy.

The 46% is relatively close to the 50% in the sample at the school in Albany.

8. There are approximately 3,289 middle students in Albany, Georgia public schools. Predict the number of students in Albany that would disagree with a mandatory uniform policy based on the data from the line plot.

$$50\% \text{ of } 3,289 = 0.5 (3,289) = 1,644.5 \text{ or } 1,645 \text{ students would disagree.}$$

9. There are approximately 3,289 middle students in Albany, Georgia public schools. Predict the number of students in Albany that would be neutral about a mandatory uniform policy based on the data from the line plot.

$$\frac{2}{30} (3,289) \approx 219 \text{ students would be neutral.}$$

10. Explain at least two factors that could contribute to the differences between your estimates and the actual number of students who could have voted about a mandatory uniform policy.

Explanations may vary.

- 1) A sample of 30 is very small for a population of 3,289 students.
- 2) The sample did not indicate how the random survey was given, which could create a bias sample.